Chapter 3: Central-Force Motion

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In this Chapter, we will consider a seemly abstract problem: the motion of a particle under the influence of a force that is always directed towards or away from a fixed point in space. This type of force is called a *central force*. Even though the problem is abstract for there actually is no such thing as a central force in nature, the problem is of great importance because it can be used to model the motion of planets and satellites under the influence of gravity, the motion of electrons around a nucleus under the influence of electrostatic forces, and many other physical phenomena.

We will begin by introducing spherical coordinates, which are the most convenient coordinate system for describing central-force motion. Then, we will define what a central force is and discuss some of its direct consequences. Finally, we will work through some examples to illustrate the concepts.

1 Spherical Coordinates

In three-dimensional space, a point can be represented in spherical coordinates (r, θ, ϕ), where r is the radial distance from the origin, θ is the polar angle measured from the positive z-axis, and ϕ is the azimuthal angle measured from the positive x-axis in the xy-plane. The relationships between Cartesian coordinates (x, y, z) and spherical coordinates are given by:

$$x = r \sin\theta \cos\phi, \tag{1}$$

$$y = r \sin\theta \sin\phi, \tag{2}$$

$$z = r \cos\theta. \tag{3}$$

2 Definition of Central Force

A force is central if:

(i) it is towards or away from a fixed point (the center of force).

(ii) the magnitude of the force is only proportional to the distance from the center of force.

Above assumptions can mathematically be expressed as

$$\mathbf{F}(\mathbf{r}) = f(r)\hat{\mathbf{r}}, \tag{4}$$

where $f(r)$ is a scalar function of the distance r from the center of force, and $\hat{\mathbf{r}}$ is the unit vector in the direction of $\mathbf{r}$.

3 Direct Consequences of the Definitions

(i) Central force cannot apply any torque on the particle. Therefore, the angular momentum of the particle is conserved.

4 Worked Examples

4.1 Work and kinetic energy in a rotating system

Two identical small masses, each of mass m, are connected to a frictionless vertical rotation axis by two massless rods of adjustable length. The system rotates in a horizontal plane with an initial angular velocity ω_i, and each mass is initially at a distance r_i from the rotation axis. The rods are then shortened, pulling the masses radially inward until their distance from the axis becomes $r_f = r_i/\sqrt{2}$. Assume that no external torque acts about the rotation axis and that the radial velocities of the masses are zero both initially and finally.

(a) Calculate the change in the total kinetic energy of the system using conservation of angular momentum.

(b) Calculate the work done by the pulling forces directly from the definition of work, $W = \int \mathbf{F} \cdot d\mathbf{r}$, and verify that it agrees with the result obtained in part (a).

Solution:

(a) Since no external torque acts about the rotation axis, the total angular momentum of the system is conserved. Treating the two masses as point particles, the initial moment of inertia is

$$I_i = 2mr_i^2. \tag{5}$$

After the masses are pulled inward to $r_f = r_i/\sqrt{2}$, the final moment of inertia becomes

$$I_f = 2mr_f^2 = 2m\left(\frac{r_i}{\sqrt{2}}\right)^2 = mr_i^2 = \frac{I_i}{2}. \tag{6}$$

From conservation of angular momentum, we have

$$I_i\omega_i = I_f\omega_f \implies 2mr_i^2\omega_i = mr_i^2\omega_f. \tag{7}$$

Therefore, the final angular velocity is

$$\boxed{\omega_f = 2\omega_i}. \tag{8}$$

Since the radial velocities are zero initially and finally, the kinetic energy at these instants is purely rotational. The initial kinetic energy is

$$K_i = \frac{1}{2}I_i\omega_i^2 = mr_i^2\omega_i^2. \tag{9}$$

Similarly, the final kinetic energy is

$$K_f = \frac{1}{2}I_f\omega_f^2 = \frac{1}{2}(mr_i^2)(2\omega_i)^2 = 2mr_i^2\omega_i^2. \tag{10}$$

Thus, the change in kinetic energy is

$$\Delta K = K_f - K_i = 2mr_i^2\omega_i^2 - mr_i^2\omega_i^2, \tag{11}$$

which gives

$$\boxed{\Delta K = mr_i^2\omega_i^2}. \tag{12}$$

The kinetic energy of the system increases even though its angular momentum remains constant. This increase must come from the work performed by the forces pulling the masses inward.

(b) To calculate this work directly, we start from the definition of work,

$$W = \int \mathbf{F} \cdot d\mathbf{r}. \tag{13}$$

Since the pulling force acts in the radial direction, only the radial component of displacement contributes to the work. For each mass, we can therefore write

$$W_1 = \int_{r_i}^{r_f} F_r\, dr, \tag{14}$$

where F_r is the radial component of the pulling force, with the outward radial direction chosen as positive.

From Newton's second law in polar coordinates, the radial acceleration of a particle is

$$a_r = \ddot{r} - r\omega^2. \tag{15}$$

Therefore, the radial force acting on each mass is

$$F_r = m(\ddot{r} - r\omega^2). \tag{16}$$

Since there are two identical masses, the total work done by the pulling forces is

$$W = 2\int_{r_i}^{r_f} F_r\, dr = 2m\int_{r_i}^{r_f} \ddot{r}\, dr - 2m\int_{r_i}^{r_f} r\omega^2\, dr. \tag{17}$$

Let us first evaluate the integral containing the radial acceleration. Using

$$\ddot{r} = \frac{d\dot{r}}{dt}, \qquad dr = \dot{r}\, dt, \tag{18}$$

we obtain

$$\int_{r_i}^{r_f} \ddot{r}\, dr = \int_{\dot{r}_i}^{\dot{r}_f} \dot{r}\, d\dot{r} = \frac{1}{2}\left(\dot{r}_f^2 - \dot{r}_i^2\right). \tag{19}$$

Since the radial velocities are zero both initially and finally, this integral vanishes. Hence, the total work reduces to

$$W = -2m\int_{r_i}^{r_f} r\omega^2\, dr. \tag{20}$$

However, the angular velocity is not constant during the inward motion. To express it as a function of the radial position r, we use conservation of angular momentum:

$$I(r)\omega(r) = I_i\omega_i. \tag{21}$$

Since $I(r) = 2mr^2$, we have

$$2mr^2\omega(r) = 2mr_i^2\omega_i, \tag{22}$$

which gives

$$\omega(r) = \frac{r_i^2}{r^2}\omega_i. \tag{23}$$

Substituting this expression into the work integral, we obtain

$$W = -2m\int_{r_i}^{r_f} r\left(\frac{r_i^2}{r^2}\omega_i\right)^2 dr \tag{24}$$

$$= -2mr_i^4\omega_i^2\int_{r_i}^{r_f}\frac{dr}{r^3}. \tag{25}$$

Evaluating the integral,

$$W = -2mr_i^4\omega_i^2\left[-\frac{1}{2r^2}\right]_{r_i}^{r_f} \tag{26}$$

$$= mr_i^4\omega_i^2\left(\frac{1}{r_f^2} - \frac{1}{r_i^2}\right). \tag{27}$$

Finally, substituting $r_f = r_i/\sqrt{2}$, we find

$$W = mr_i^4\omega_i^2\left(\frac{2}{r_i^2} - \frac{1}{r_i^2}\right) \tag{28}$$

$$= mr_i^2\omega_i^2. \tag{29}$$

Therefore,

$$\boxed{W = mr_i^2\omega_i^2}. \tag{30}$$

This result agrees exactly with the change in kinetic energy calculated in part (a):

$$\boxed{W = \Delta K = K_f - K_i}. \tag{31}$$

Thus, although the angular momentum of the system is conserved, its kinetic energy increases due to the positive work performed by the pulling forces. This demonstrates that conservation of angular momentum does not necessarily imply conservation of kinetic energy.